Algebraic Geometry Cheat Sheet

Dmytro Zakharov

Contents

1 Definitions **2**
- 1.1 Affine and Projective Sets . . . 2
- 1.2 Some Topology Facts . . . 3
- 1.3 Separated and complete algebraic sets . . . 3
- 1.4 Function Fields, Rational Maps . . . 4
- 1.5 Smooth Varieties . . . 5
- 1.6 Locally ringed spaces and schemes . . . 5
- 1.7 Properties of Schemes and Morphisms of Schemes . . . 6
- 1.8 Coherent sheaves, Picard Group . . . 7

1 Definitions

1.1 Affine and Projective Sets

Further assume $k = \overline{k}$ (*i.e.*, k is algebraically closed).

Definition 1.1. *We define affine and projective algebraic sets as follows:*

- *An **affine algebraic set** is a subset $Z \subseteq k^n$ of the form $V(I)$ for $I \subseteq k[T_1, \ldots, T_n]$.*
- *A **projective algebraic set** is a subset $Z \subseteq \mathbb{P}^n(k)$ of form $V_{\mathbb{P}}(I)$ for homogeneous ideal $I \subseteq k[T_1, \ldots, T_n]$ where $\mathbb{P}^n(k) = (k^{n+1} \setminus 0)/k^\times$.*

Definition 1.2. *Let $X \subseteq k^n$ be an algebraic set, $U \subseteq X$ an open subset, and let $f : U \to k$ be a function. We say that f is a **regular function** on U if*

$$\forall_{x \in U}\ \exists_{x \ni V \subseteq U}\ \exists_{p,q \in k[T_1,\ldots,T_n]}\ \forall_{y \in V}\ q(y) \neq 0 \text{ and } f(y) = p(y)/q(y).$$

Definition 1.3. *A **space with functions** (swf) is a topological space X together with an assigment, for every open $U \subseteq X$, of a k-subalgebra $\mathcal{O}(U)$ of the ring of all functions $U \to k$ such that:*

(a) Being regular is a local propery. That is, if $U = \bigcup_{\alpha \in I} U_\alpha$ is an open cover and $f : U \to k$ is a function, then $f \in \mathcal{O}(U)$ iff $f|_{U_\alpha} \in \mathcal{O}(U_\alpha)$ for each $\alpha \in I$.

(b) If $U \subseteq X$ is an open subset and $f \in \mathcal{O}(U)$, then the set $D(f) = \{x \in U : f(x) \neq 0\} \subseteq U$ is an open subset of U and $(f|_{D(f)})^{-1} \in \mathcal{O}(D(f))$.

*A **morphism of swf's** is a continuous map $\phi : Y \to X$ such that the pullbacks of regular functions are regular: for every open $U \subseteq X$ and every regular function $f \in \mathcal{O}(U)$, the function $f \circ \phi \in \mathcal{O}(\phi^{-1}(U))$. Note that ϕ induces k-algebra homomorphism $\phi^* : \mathcal{O}(X) \to \mathcal{O}(Y)$.*

Example. *Let $k = \mathbb{R}$ or $k = \mathbb{C}$ and let X be a topological space. Then,*

$$\mathcal{O}(U) = C(U; k) = \{f : U \to k, f \text{ is continuous}\}$$

gives X a structure of a space with functions.

Definition 1.4. *An **algebraic set** X over k is a space with functions X admitting a finite open cover $X = U_1 \cup \cdots \cup U_n$ where each U_i is isomorphic to an swf to an affine algebraic set. We call X a **variety** if it is irreducible (i.e., if for every pair of closed subsets $Y_1, Y_2 \subseteq X$ with $X = Y_1 \cup Y_2$ we have $X = Y_1$ or $X = Y_2$).*

Example. *A couple of examples:*

- *The projective space $\mathbb{P}^n$ has the standard cover $U_0 \cup \cdots \cup U_n$ by affine spaces. Every locally closed subset (i.e. closed subset Z of $\mathbb{P}^n$ such that $Z \cap U_i$ are regular) of $\mathbb{P}^n$ is an swf and an algebraic set.*
- *The punctured plane $\mathbb{A}^2 \setminus 0$ is not affine.*

1.2 Some Topology Facts

Definition 1.5. *For a topological space X, the **dimension** $\dim(X)$ is the supremum of the set of integers $n \geq 0$ for which there exists a chain of distinct closed irreducible subsets $Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n$.*

Definition 1.6. *We say that X is **Noetherian** if every decreasing sequence $F_0 \supseteq F_1 \supseteq \ldots$ of closed subsets stabilizes.*

Lemma 1.7. *Let X be a Noetherian topological space. Then there exist closed irreducible subsets $Z_1, \ldots, Z_r \subseteq X$ such that $Z_i \not\subseteq Z_j$ for $i \neq j$ and $X = Z_1 \cup \cdots \cup Z_r$. They are unique up to permutation.*

Definition 1.8. *The closed irreducible subsets $Z_1, \ldots, Z_r \subseteq X$ in the lemma are called the **irreducible components** of X.*

1.3 Separated and complete algebraic sets

Definition 1.9. *A morphism of swf's $Y \to X$ is an **immersion** if it is a homeomorphism onto its image and the swf structure on Y coincides on the image. A **closed** (resp. **open**, resp. **locally closed**) immersion is an immersion whose image is closed (resp. open, resp. locally closed) in X.*

Example. *Consider the map $f : \mathbb{A}^1 \to \mathbb{A}^2, t \mapsto (t^2, t^3)$. It is a homeomorphism onto its image, which is the cuspidal curve $C = V(Y^2 - X^3)$. However, it is not an immersion, since the map on coordinate rings (note that coordinates rings are given by $\mathcal{O}(C) = k[X,Y]/(Y^2 - X^3)$ and $\mathcal{O}(\mathbb{A}^1) = k[T]$) is*

$$f^* : k[X,Y]/(Y^2 - X^3) \to k[T], \quad X \mapsto T^2, Y \mapsto T^3$$

is not an isomorphism.

Definition 1.10. *An algebraic set X is **separated** if the diagonal $\Delta := \{(x,x) : x \in X\} \subseteq X \times X$ is a closed subset of $X \times X$.*

Definition 1.11. *An algebraic set X is **complete** if X is separated and for every algebraic set Y, the projection map $\pi_Y : X \times Y \to Y$ is closed.*

Example. *A couple of examples:*

- *If X is affine, then X is separated. Indeed, exists $X \hookrightarrow \mathbb{A}^n$ and $\mathbb{A}^n$ is separated, as its diagonal in $\mathbb{A}^n \times \mathbb{A}^n$ (with coordinates $X_1, \dots, X_n, Y_1, \dots, Y_n$) is cut out by equations $X_i = Y_i$ and hence is closed. However, $\mathbb{A}^n$ is **not** complete: take $Y := \mathbb{A}^1$ with coordinate T and the closed subset $Z = V(fT - 1) \subseteq X \times Y$. Its projection onto Y is then an infinite subset which does not contain zero, and hence cannot be closed.*
- *The projective space $\mathbb{P}^n$ is both separated (one sees this using Segre embedding $\mathbb{P}^n \times \mathbb{P}^n \to \mathbb{P}^{n^2+2n}$) and complete. Moreover, any projective algebraic set is complete.*

1.4 Function Fields, Rational Maps

Definition 1.12. *Let X be an algebraic set and let $Z \subseteq X$ be an irreducible closed subset. We define the **stalk at** Z (also called the **local ring** at Z) as the filtered colimit*

$$\mathcal{O}_{X,Z} = \varinjlim_{U \cap Z \neq \emptyset} \mathcal{O}(U).$$

*The ring $\mathcal{O}_{X,\{x\}}$, further simply denoted as $\mathcal{O}_{X,x}$ is called a **germ** of regular functions in a neighbourhood of x. The ring $\mathcal{O}_{X,X}$ (when X is a variety), denoted by $k(X)$, is called a **function field** of X.*

Example. *Let $X = V(z^2 - xy) \subseteq \mathbb{A}^3$. Let $x_0 = (1,1,1) \in X$. We look at the stalk $\mathcal{O}_{X,x_0}$. Consider two functions:*

- *$f = x/z$ which is defined on open $U = D(z)$, $x_0 \in U$;*
- *$g = z/y$ which is defined on open $V = D(y)$, $x_0 \in V$;*

then (U, f) and (V, g) are the same elements in $\mathcal{O}_{X,x_0}$. Indeed, at intersection $U \cap V = D(YZ)$ one notices $x/z = z/y$ and so $f|_{U \cap V} = g|_{U \cap V}$.

Definition 1.13. *Let X and Y be varieties. A **rational map** from Y to X is an equivalence class of pairs (U, f) where $U \subseteq Y$ is a non-empty open subset and where $f : U \to X$ is a map of varieties, where we identify (U, f) and (U', f') if $f = f'$ on some non-empty open $U'' \subseteq U \cap U'$. We call a rational map (U, f) **dominant** if $f(U)$ is dense in X.*

Example. *A couple of examples:*

- *Let* $Y = V(x^2 + y^2 - 1) \subseteq \mathbb{A}^2$ *and let* $X = \mathbb{A}^1$. *Define two rational maps:*

$$f : Y \dashrightarrow X,\ (x,y) \mapsto x/(1-y),\ U = Y \setminus \{(0,1)\},$$
$$g : Y \dashrightarrow X,\ (x,y) \mapsto (1+y)/x,\ V = Y \setminus \{(0,\pm 1)\}.$$

  One checks that (U, f) *and* (V, g) *define the same rational map.*
- *Let* $Y = \mathbb{P}^2$ *and* $X = \mathbb{P}^1$. *Define* $g : Y \to X$ *as* $[x : y : z] \mapsto [x : y]$. *The map is defined on the open set* $U = \mathbb{P}^2 \setminus \{[0 : 0 : 1]\}$. *It is a dominant map.*
- *Let* $Y = \mathbb{A}^1$ *and* $X = \mathbb{A}^2$. *Define* $h : Y \dashrightarrow X$ *by* $t \mapsto (t, 1/t)$. *The map is defined on open* $\mathbb{A}^1 \setminus \{0\}$ *and its image is* $V(xy - 1) \subseteq \mathbb{A}^2$. *Clearly it is not dominant as the image is closed proper subset of* $\mathbb{A}^2$.

Definition 1.14. *A rational map* $f : Y \dashrightarrow X$ *is* ***birational*** *if it is dominant and if it admits an inverse, or equivalently if there exist non-empty opens* $V \subseteq Y$ *and* $U \subseteq X$ *such that* f *induces an isomorphism* $V \cong U$ *(or* $k(X) \cong k(Y)$*.) We say that* X *and* Y *are* ***birational*** *if there exists a birational rational map from* Y *to* X. *We say that a variety* X *is* ***rational*** *if it is birational to* $\mathbb{P}^n$ *for some* $n \geq 0$.

Example. *A couple of examples:*

- *Let* $X = V(f) \subseteq \mathbb{P}^3$ *be an irreducible quadric surface (so* $f \in k[T_0, \ldots, T_3]$ *is an irreducible homogeneous polynomial of degree two). Then,* X *is rational.*
- *The cubic plane curve* $V(y^2 - x^3 - x)$ *is not rational.*

1.5 Smooth Varieties

Definition 1.15. *Let* X *be an algebraic set and let* $x \in X$. *We say that* X *is* ***nonsingular*** *at* x *if* $\mathcal{O}_{X,x}$ *is a regular local ring. If every point is nonsingular, then* X *is called* ***smooth****.*

Definition 1.16. *Let* A *be a finitely generated and reduced* k*-algebra. Let* $x \in MSpec(A) = Hom_k(A, k)$ *be a point of the corresponding affine algebraic set. The* ***Zariski cotangent space*** *at* x *is the vector space over* k

$$T_x = (\mathfrak{m}_x/\mathfrak{m}_x^2)^\vee = Hom_k(\mathfrak{m}_x/\mathfrak{m}_x^2, k).$$

1.6 Locally ringed spaces and schemes

Definition 1.17.

- *A* ***ringed space*** *is a pair* $(X, \mathcal{O}_X)$ *where* X *is a topological space and* $\mathcal{O}_X$ *is a sheaf of rings on* X.
- *A* ***locally ringed space*** *is a ringed space* $(X, \mathcal{O}_X)$ *such that for each* $x \in X$, *the stalk* $\mathcal{O}_{X,x}$ *is a local ring.*

Definition 1.18.
- *An **affine scheme** is a locally ringed space $(X, \mathscr{O}_X)$ isomorphic to $(Spec(A), \mathscr{O}_{Spec(A)})$ for some ring A (note that then $A = \mathscr{O}_X(X)$).*
- *A **scheme** is a locally ringed space $(X, \mathscr{O}_X)$ which is locally isomorphic to an affine scheme; i.e., there exists an open cover $X = \bigcup_{\alpha \in I} U_\alpha$ such that each $(U_\alpha, \mathscr{O}_X|_{U_\alpha})$ is an affine scheme.*
- *A **morphism of ringed spaces** $(Y, \mathscr{O}_Y) \to (X, \mathscr{O}_X)$ is a pair (ϕ, ϕ^*) where $\phi : Y \to X$ is a continuous map and where $\phi^* : \mathscr{O}_X \to \phi_* \mathscr{O}_Y$ is a homomorphism of sheaves of rings on X.*
- *A **morphism of locally ringed spaces** $(Y, \mathscr{O}_Y) \to (X, \mathscr{O}_X)$ is a map of ringed spaces $\phi : Y \to X$ such that for every $y \in Y$, the induced map $\phi^* : \mathscr{O}_{X,\phi(x)} \to \mathscr{O}_{Y,y}$ is a local ring homomorphism (maps maximal ideals into maximal ideals).*

1.7 Properties of Schemes and Morphisms of Schemes

Definition 1.19. *Let X be a scheme.*

*(a) We say that X is **reduced** if $\mathscr{O}_X(U)$ is reduced for every open $U \subseteq X$, and that X is **integral** if it is reduced and irreducible (equivalently, $\mathscr{O}_X(U)$ is a domain for every non-empty open $U \subseteq X$).*

*(b) We say that X is **noetherian** if it admits a finite open cover $X = U_1 \cup \cdots \cup U_r$ with $U_i \cong Spec(A_i)$ where the rings $A_1, \ldots, A_r$ are noetherian.*

Definition 1.20 (Morphisms of schemes)**.** *Let $f : Y \to X$ be a morphism of schemes. We say that f is*

*(a) **affine** is for every affine open $U \subseteq X$, the preimage $f^{-1}(U) \subseteq Y$ is affine;*

*(b) **open immersion** if it induces an isomorphism from Y onto the open subscheme of X;*

*(c) **closed immersion** if it induces a homeomorphism from Y onto a closed subset of X and the map $\mathscr{O}_X \to f_* \mathscr{O}_Y$ is surjective;*

*(d) **finite** if it is affine and for every affine open $U \subseteq X$, the map $\mathscr{O}_X(U) \to \mathscr{O}_Y(f^{-1}(U))$ is finite;*

*(e) **locally of finite type** for every $y \in Y$ there exist affine open neighbourhoods $y \in V$ and $f(x) \in U$ such that $f(V) \subseteq U$ and the map $f^* : \mathscr{O}_X(U) \to \mathscr{O}_Y(V)$ is of finite type;*

*(f) **of finite type** if it is locally of finite type and quasi-compact.*

Example. *A couple of illustrating examples:*

(a) Let $Y = \mathbb{A}^3(k)$ and $X = \mathbb{A}^2(k)$ with the projection map $(x, y, z) \mapsto (x, y)$. This induces the homomorphism of shaves of rings $k[x, y] \to k[x, y, z]$, which is affine.

(b) Let $Y = X \setminus \{0\}$ and $X = \mathbb{A}^1(k)$ with inclusion map. This map is an open immersion for obvious reasons.

(c) Let $Y = V(y - x^2)$ and $X = \mathbb{A}^2(k)$. The morphism $f : Y \to X, (x, y) \mapsto x$ induces a morphism $k[x] \to k[x, y]/(y^2 - x)$ which is a closed immersion.

1.8 Coherent sheaves, Picard Group

Definition 1.21. *Let $\mathscr{O}$ be a sheaf of rings on a topological space X. An $\mathscr{O}$-**module** is a sheaf of abelian groups $\mathscr{M}$ together with a map of sheaves $\mathscr{O} \times \mathscr{M} \to \mathscr{M}$ giving $\mathscr{M}(U)$ the structure of an $\mathscr{O}(U)$-module for every open $U \subseteq X$.*

Example. *If X is a C^∞-manifold and $\mathscr{O}$ is the sheaf of smooth real-valued functions on X, the sheaf $\mathscr{M}$ of differential 1-forms on X has a natural structure of an $\mathscr{O}$-module.*

Example. *Let $X = Spec(A)$ be an affine scheme and let M be an A-module. We can turn M into an $\mathscr{O}_X$-module $\widetilde{M}$ such that $\widetilde{M}(D(f)) = M[f^{-1}]$.*

Definition 1.22. *Let X be either a scheme or an algebraic set. A $\mathscr{O}_X$-module $\mathscr{F}$ is **quasi-coherent** if there exists an affine open cover $X = \bigcup_{\alpha \in I} U_\alpha$ such that $\mathscr{F}|_{U_\alpha}$ is of the form $\widetilde{M}_\alpha$ for some $\mathscr{O}(U_\alpha)$-module M_α. If X is either an algebraic set or locally noetherian, we say that $\mathscr{F}$ is **coherent** if same holds for M_α being finitely generated $\mathscr{O}(U_\alpha)$-module.*